



Basic Tree Risk Assessment Form

Client Field test 11 Date 04/09/2026 Time 11:52
 Address/Tree location Arboretum, CSUS Tree no. 1 Sheet of
 Tree species red horse chestnut dbh 12" Height 34' Crown spread dia. 30'
 Assessor(s) Gary Hare Time frame Tools used

Target Assessment

Target number	Target description	Target zone			Occupancy rate 1 – rare 2 – occasional 3 – frequent 4 – constant	Practical to move target?	Restriction practical?
		Target within drip line	Target within 1 x Ht.	Target within 1.5 x Ht.			
<u>1</u>	<u>Pathway</u>	<u>X</u>	<u>X</u>	<u> </u>	<u>2</u>	<u> </u>	<u>X</u>
<u>2</u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>
<u>3</u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>
<u>4</u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>	<u> </u>

Site Factors

History of failures No history of failures Topography Flat ☒ Slope ☐ % Aspect
 Site changes None ☐ Grade change ☐ Site clearing ☐ Changed soil hydrology ☐ Root cuts ☐ Describe flat grove
 Soil conditions Limited volume ☐ Saturated ☐ Shallow ☒ Compacted ☐ Pavement over roots ☐ % Describe Shallow, moist but not sat
 Prevailing wind direction NW Common weather Strong winds ☐ Ice ☐ Snow ☐ Heavy rain ☐ Describe Medium winds, occasional rain

Tree Health and Species Profile

Vigor Low ☐ Normal ☒ High ☐ Foliage None (seasonal) ☐ None (dead) ☐ Normal 85 % Chlorotic % Necrotic %
 Pests No pests observed Abiotic No abiotic stress observed
 Species failure profile Branches ☒ Trunk ☒ Roots ☐ Describe Several hanging branches, potential for branch failures.

Load Factors

Wind exposure Protected ☐ Partial ☒ Full ☐ Wind funneling ☐ Relative crown size Small ☒ Medium ☐ Large ☐
 Crown density Sparse ☒ Normal ☐ Dense ☐ Interior branches Few ☒ Normal ☐ Dense ☐ Vines/Mistletoe/Moss ☐ No vines/mistletoe/moss observed
 Recent or planned change in load factors none noted

Tree Defects and Conditions Affecting the Likelihood of Failure

— Crown and Branches —

Unbalanced crown ☒ LCR 60 %
 Dead twigs/branches ☒ 15 % overall Max. dia. 3"
 Broken/Hangers Number 5 Max. dia. 2"
 Over-extended branches ☒
 Pruning history
 Crown cleaned ☐ Thinned ☐ Raised ☐
 Reduced ☐ Topped ☐ Lion-tailed ☐
 Flush cuts ☐ Other
 Cracks ☐ Lightning damage ☐
 Codominant ☒ Included bark ☒
 Weak attachments ☒ Cavity/Nest hole 8 % circ.
 Previous branch failures ☐ Similar branches present ☐
 Dead/Missing bark ☒ Cankers/Galls/Burls ☐ Sapwood damage/decay ☒
 Conks ☐ Heartwood decay ☒ present in a major scaffold branch
 Response growth none
 Main concern(s) the decay in the scaffold branch in the crown and the broken hangers coming from in the crown

Load on defect N/A ☐ Minor ☐ Moderate ☒ Significant ☐ moderate
 Likelihood of failure Improbable ☐ Possible ☒ Probable ☐ Imminent ☐ possible

— Trunk —

Dead/Missing bark ☒ Abnormal bark texture/color ☒
 Codominant stems ☒ Included bark ☐ Cracks ☐
 Sapwood damage/decay ☒ Cankers/Galls/Burls ☐ Sap ooze ☐
 Lightning damage ☐ Heartwood decay ☒ Conks/Mushrooms ☐
 Cavity/Nest hole 50 % circ. Depth 2" Poor taper ☐
 Lean ° Corrected? look it's going like that
 Response growth none
 Main concern(s) there are two cavities on the trunk

Load on defect N/A ☐ Minor ☐ Moderate ☒ Significant ☐
 Likelihood of failure Improbable ☐ Possible ☒ Probable ☐ Imminent ☐

— Roots and Root Collar —

Collar buried/Not visible ☒ Depth 4" Stem girdling ☒
 Dead ☐ Decay ☐ Conks/Mushrooms ☐
 Ooze ☐ Cavity ☐ % circ.
 Cracks ☐ Cut/Damaged roots ☐ Distance from trunk
 Root plate lifting ☐ Soil weakness ☐
 Response growth none
 Main concern(s) the roots are buried under fairly compacted soil and there may be stem girdling going on

Load on defect N/A ☒ Minor ☐ Moderate ☐ Significant ☐
 Likelihood of failure Improbable ☒ Possible ☐ Probable ☐ Imminent ☐

Risk Categorization																											
Condition number	Tree part	Conditions of concern	Part size	Fall distance	Target number	Target protection	Likelihood												Consequences				Risk rating of part (from Matrix 2)				
							Failure				Impact				Failure & Impact (from Matrix 1)												
							Improbable	Possible	Probable	Imminent	Very low	Low	Medium	High	Unlikely	Somewhat	Likely	Very likely	Negligible	Minor	Significant	Severe					
1	Crown	Dead branching cavity	4	14 to 20'	1			X				X			X								X		low		
2	Trunk		8	20'	1	none		X					X		X								X		low		
2																											
3																											
4																											

Matrix 1. Likelihood matrix.

Likelihood of Failure	Likelihood of Impacting Target			
	Very low	Low	Medium	High
Imminent	Unlikely	Somewhat likely	Likely	Very likely
Probable	Unlikely	Unlikely	Somewhat likely	Likely
Possible	Unlikely	Unlikely	Unlikely	Somewhat likely
Improbable	Unlikely	Unlikely	Unlikely	Unlikely

Matrix 2. Risk rating matrix.

Likelihood of Failure & Impact	Consequences of Failure			
	Negligible	Minor	Significant	Severe
Very likely	Low	Moderate	High	Extreme
Likely	Low	Moderate	High	High
Somewhat likely	Low	Low	Moderate	Moderate
Unlikely	Low	Low	Low	Low

Notes, explanations, descriptions The tree is a red horse chestnut at CSUS Arboretum. It has a nesting cavity, an unbalanced crown, and a rotten scaffolding branch. Roots are buried, and the soil is compacted. Growth shows no necrosis or chlorosis.

Mitigation options keep pruning and remove the dead branch

Residual risk low

Residual risk _____

Residual risk _____

Residual risk _____

Overall tree risk rating Low ☒ Moderate ☐ High ☐ Extreme ☐

Work priority 1 ☐ 2 ☐ 3 ☐ 4 ☐

Overall residual risk Low ☒ Moderate ☐ High ☐ Extreme ☐

Recommended inspection interval 1 year

Data ☐ Final ☐ Preliminary Advanced assessment needed ☐ No ☐ Yes-Type/Reason _____

Inspection limitations ☐ None ☐ Visibility ☐ Access ☐ Vines ☐ Root collar buried Describe _____

